A foredune built by marram grass

Simon Frost

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1 Overview

With `veg_model = marram`, every grid cell carries the marram grass model of the previous vignette, and the plants and the bed are coupled both ways:

- Plants shape the wind. Each time step, shoot density and canopy height give a frontal area. That frontal area becomes the effective cover in AeoLiS's shear reduction, `1 / sqrt(1 + gamma_vegshear rhoveg)`. The shoots bend in strong wind, so the cover depends on the current wind speed (Laporte-Fauret et al. 2024; Walter et al. 2012).
- The bed shapes the plants. Every `veg_interval` (a day), each cell is advanced using:
  - the rate of bed-level change since the last update, as burial;
  - the fraction of the interval it spent under water;
  - rainfall and evapotranspiration from `veg_water_file`;
  - the day of the year from `refdate`.
- Plants spread. Bare cells are colonised from occupied neighbours at about 0.4 m per year (Keijsers et al. 2015), with occasional jumps (Bonte et al. 2021).

- Roots resist waves. When a storm cuts back the dune, sand in the rooted layer costs more of the waves' work to remove than bare sand (Figlus et al. 2014; De Battisti and Griffin 2020).

```
ENV["GKSwstype"] = "100"
using Aeolis
using Plots
using Printf
using Random
```

2 The beach and its weather

A transect rises from below the low-tide line to a backshore at 3 m, 100 m inland. Beyond 126 m the hinterland is fixed, so the bare sand behind the dune does not blow away. There is a semidiurnal tide of 1 m. Every day has a new onshore wind: a Weibull speed with a mean of 7 m/s, from directions within 40° of shore-normal. Rain falls on 45 % of days, and potential evapotranspiration follows the seasons. The weather is drawn from a seeded random number generator, so the vignette is reproducible.

```
day = 86400.0
years = 3
rng = Xoshiro(1)
x = collect(0.0:1.0:150.0)
zb0 = @. min(-1.0 + x / 25, 3.0)
nd = 365 * years + 1
td = (0:(nd - 1)) .* day
wind = hcat(td, [7.0 / 0.886 * sqrt(-log(rand(rng))) for _ in td],
            [270.0 + 80 * (rand(rng) - 0.5) for _ in td])
tt = collect(0.0:3600.0:(nd * day))
tide = hcat(tt, sin.(2pi .* tt ./ (12.42 * 3600)))
doy = [(k % 365) + 1 for k in 0:(nd - 1)]
water = hcat(td, [rand(rng) < 0.45 ? -4.9 * log(rand(rng)) : 0.0 for _ in td],
             @. 2.0 - 1.5 * cos(2pi * (doy - 15) / 365))
base = (; xgrid_file=x, bed_file=zb0, ne_file=(@. ifelse(x >= 126, zb0, -10.0)),
        th_nelayer=true, wind_file=wind, tide_file=tide, process_tide=true,
        grain_size=0.0003, dt=3600.0, tstop=years * 365day, output_times=30day,
        refdate="2020-01-01 00:00", process_avalanche=true,
        output_vars=[:zb, :veg_N, :hveg, :rhoveg, :veg_P, :veg_Gb, :veg_zdot])
nothing
```

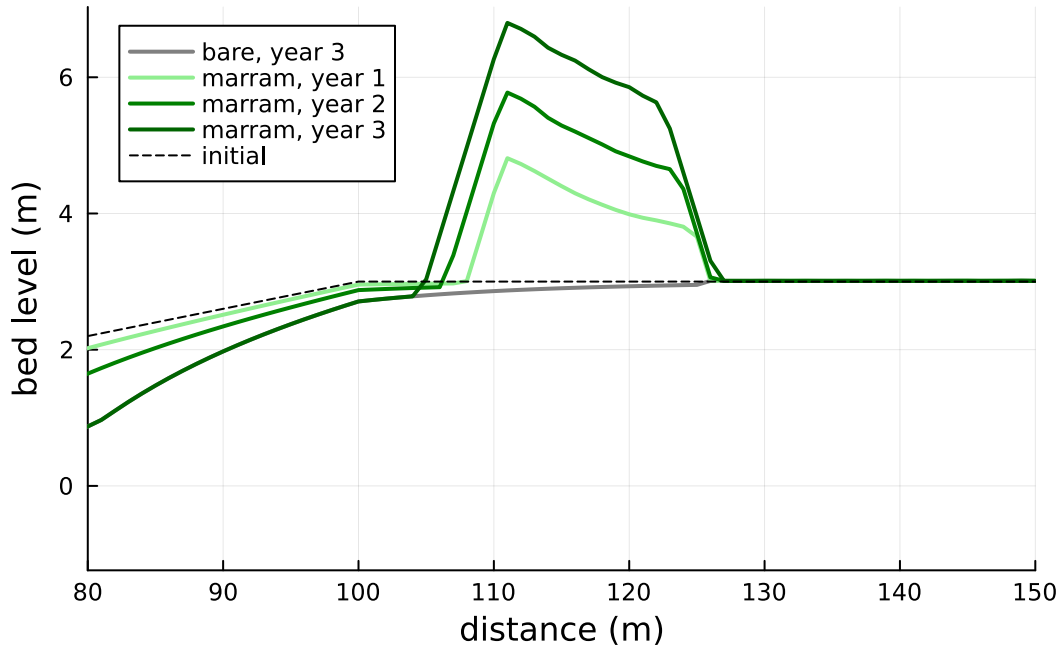
Marram is planted between 110 and 125 m at 30 % of the density of a closed stand, 141 shoots per m^2. The same beach is also run without plants.

```
planting = @. ifelse(110 <= x <= 125, 0.3, 0.0)
marram = solve(AeolisProblem(AeolisParameters(; base..., process_vegetation=true,
                                                veg_model=:marram, veg_file=planting,
```

```
                                                veg_water_file=water));
rng=Xoshiro(2))
bare = solve(AeolisProblem(AeolisParameters(; base...)))
nothing
```

3 The dune

```
z = transect(marram, :zb)
zbare = transect(bare, :zb)
volume(zz) = sum(max.(zz[101:end] .- 3.0, 0.0))     # m³ per m of beach, above 3 m
plt = plot(x, zbare[:, end]; color=:gray, lw=2, label="bare, year 3",
           xlabel="distance (m)", ylabel="bed level (m)", xlims=(80, 150),
legend=:topleft)
for (k, c) in zip((13, 25, 37), (:lightgreen, :green, :darkgreen))
    plot!(plt, x, z[:, k]; lw=2, color=c, label="marram, year $((k - 1) ÷ 12)")
end
plot!(plt, x, zb0; color=:black, ls=:dash, label="initial")
plt
```



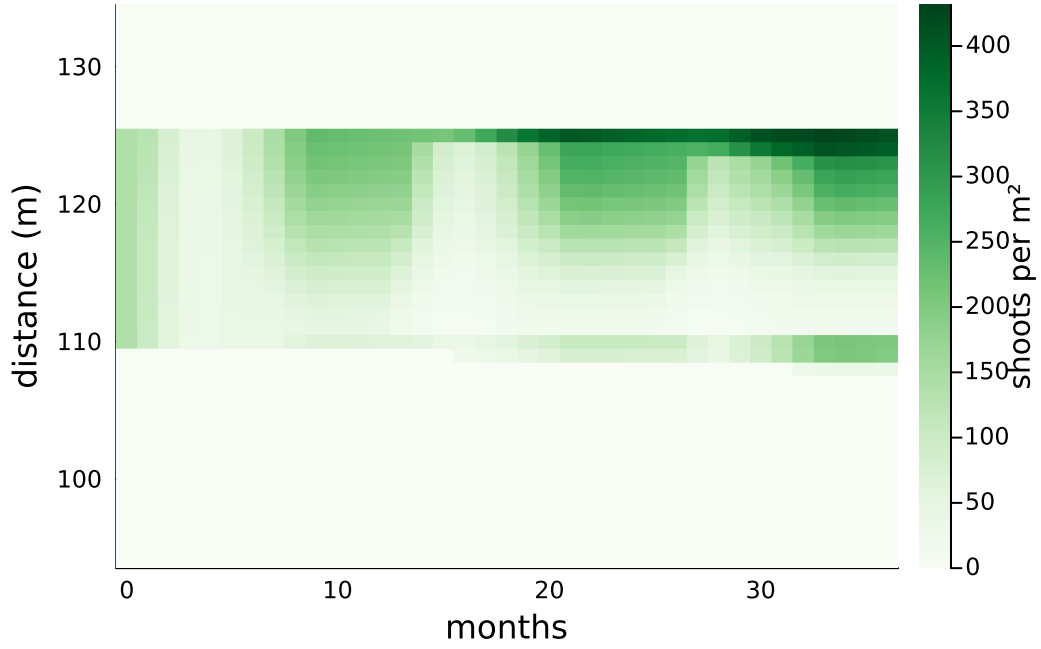

```
for k in (13, 25, 37)
    @printf("year %d: dune volume %.1f m³/m with marram, %.1f m³/m bare; crest %.2f
    m\n",
            (k - 1) ÷ 12, volume(z[:, k]), volume(zbare[:, k]), maximum(z[:, k]))
end
```

```
year 1: dune volume 20.1 m³/m with marram, 0.3 m³/m bare; crest 4.81 m
year 2: dune volume 35.7 m³/m with marram, 0.3 m³/m bare; crest 5.78 m
year 3: dune volume 53.6 m³/m with marram, 0.3 m³/m bare; crest 6.80 m
```

The bare beach builds no dune: sand blown off the beach crosses the flat backshore. The planted strip slows the wind at the bed, traps the sand and grows a dune several metres high in three years. The wind blows onshore every day here, so this rate is high for a real coast.

4 The plants

```
N = transect(marram, :veg_N)
months = (0:(size(N, 2) - 1))
heatmap(months, x[95:135], N[95:135, :]; xlabel="months", ylabel="distance (m)",
        colorbar_title="shoots per m²", c=:Greens)
```



Shoot density rises each summer and falls back each winter.

- Seaward edge: the stand spreads over the bare beach a little each year, which is as much as the front speed allows.
- Crest: the stand thins where the dune grows fastest.

The diagnostics at the end of the run show why:

```
G = transect(marram, :veg_Gb)
zdot = transect(marram, :veg_zdot)
tr = MarramTraits()
for i in 106:4:126
    @printf("x = %3d m: bed %.2f m, burial %5.2f m/season, G_b %.2f, shoots %3.0f
    per m²\n",
            x[i], z[i, end], zdot[i, end] * Aeolis.season_length(tr), G[i, end],
N[i, end])
end
```

```
x = 105 m: bed 3.03 m, burial  0.47 m/season, G_b 0.82, shoots   0 per m²
x = 109 m: bed 5.62 m, burial  0.50 m/season, G_b 0.77, shoots 199 per m²
x = 113 m: bed 6.60 m, burial  0.72 m/season, G_b 0.30, shoots  30 per m²
x = 117 m: bed 6.12 m, burial  0.50 m/season, G_b 0.77, shoots 126 per m²
x = 121 m: bed 5.73 m, burial  0.34 m/season, G_b 0.99, shoots 259 per m²
x = 125 m: bed 3.95 m, burial -0.04 m/season, G_b 0.98, shoots 414 per m²
```

Where the crest is buried by about 0.7 m per growing season, more than twice the optimum (Nolet et al. 2018), the burial multiplier `G_b` falls to about 0.3. Fewer shoots survive there than on the flanks. The landward edge receives little sand. The model would slowly lower vigour there through soil-biota pressure, but three years is too short for that to show.

5 How densely to plant

The stand is planted at densities from 2 % to 100 % of a closed stand, and run for two years:

```
for c in (0.02, 0.1, 1.0)
    p = AeolisParameters(; base..., tstop=2 * 365day, output_times=365day,
                         process_vegetation=true, veg_model=:marram,
                         veg_file=(@. ifelse(110 <= x <= 125, c, 0.0)),
                         veg_water_file=water)
    zz = transect(solve(AeolisProblem(p); rng=Xoshiro(2)), :zb)
    @printf("planted at %4.0f %%: dune volume %.1f m³/m after two years\n", 100c,
            volume(zz[:, end]))
end
```

```
planted at    2 %: dune volume 33.1 m³/m after two years
planted at   10 %: dune volume 34.4 m³/m after two years
planted at  100 %: dune volume 38.0 m³/m after two years
```

Even a sparse planting builds most of the dune, for two reasons:

- Shelter saturates. The drag on full-height marram reaches its cap at about 45 shoots per m².
- Sparse stands fill in. They grow toward their capacity within a season or two.

Bonte et al. (2021) found that dunes grow fastest at intermediate cover. The model does not reproduce this, either here or in two-dimensional runs with patchy planting, where dune volume also rose with the area planted. Why the two differ is not yet understood. The supply of sand, the measure of dune growth, and how trapping depends on cover in their cellular model are candidates.

6 A storm

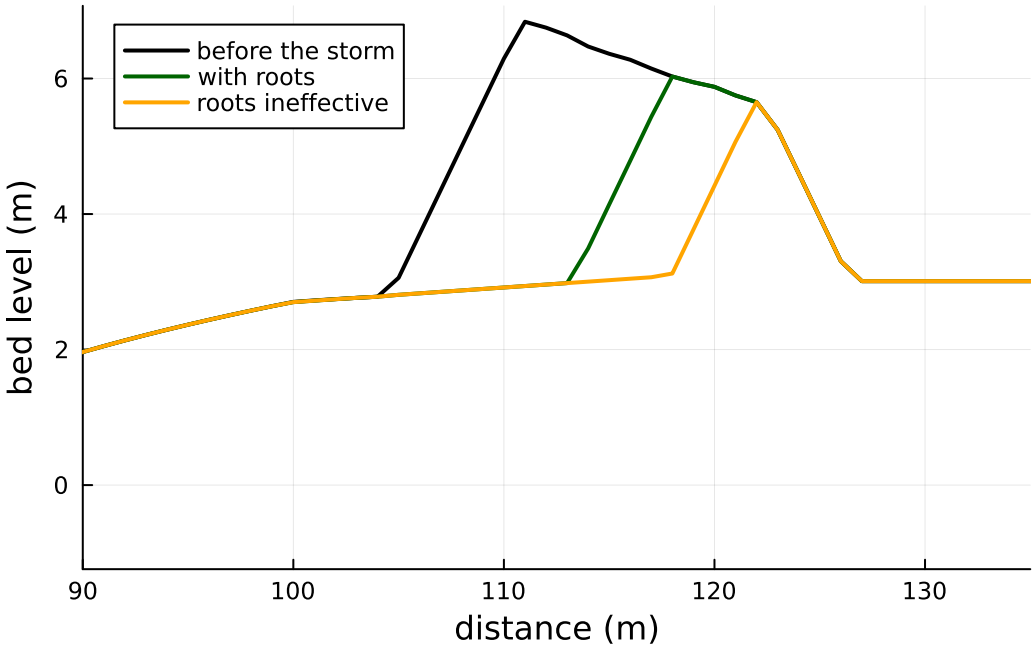
After the three years, a storm attacks the dune for two days. A surge of 2.5 m and waves 3 m high with a 10 s period collide with the dune face, which erodes as in Palmsten and Holman (2012).

The storm run starts from the state after three years (`u0`), so the roots are those the dune grew. It is repeated with the roots made ineffective: a traits file sets the strength of bare sand, `tau0`, so high that reinforcement adds nothing. The light wind moves no sand.

The dune toe is set at 2.7 m. AeoLiS takes as the toe the point of the profile nearest that elevation, and the hinterland lies at exactly 3 m.

```
t0 = years * 365day
storm = (; tstart=t0, tstop=t0 + 2day, output_times=2day,
         wind_file=[0.0 2.0 270.0; 4000day 2.0 270.0], tide_file=[0.0 2.5; 4000day
2.5],
         wave_file=[0.0 3.0 10.0; 4000day 3.0 10.0], process_wave=true,
process_runup=true,
         process_dune_erosion=true, dune_toe_elevation=2.7, beach_slope=0.04)
inert = tempname() * ".toml"
write(inert, "tau0 = 1e12\n")
u3 = marram.u
eroded = Dict{String,Float64}()
plt = plot(x, u3.zb[:, 2]; color=:black, lw=2, label="before the storm",
           xlabel="distance (m)", ylabel="bed level (m)", xlims=(90, 135),
legend=:topleft)
for (label, kw, c) in (("with roots", (;), :darkgreen), ("roots ineffective", (;
veg_traits_file=inert), :orange))
    p = AeolisParameters(; base..., storm..., process_vegetation=true,
veg_model=:marram,
                         veg_file=planting, kw...)
    z1 = solve(AeolisProblem(p; u0=u3); rng=Xoshiro(3)).u.zb[:, 2]
    eroded[label] = sum(max.(u3.zb[:, 2] .- z1, 0.0))
    plot!(plt, x, z1; lw=2, color=c, label=label)
end
@printf("eroded: %.1f m³/m with roots, %.1f m³/m without; %.0f %% less\n",
        eroded["with roots"], eroded["roots ineffective"],
        100 * (1 - eroded["with roots"] / eroded["roots ineffective"]))
plt
```

```
eroded: 29.7 m³/m with roots, 42.6 m³/m without; 30 % less
```



The roots keep a third of the dune face that the storm would otherwise take. Flume experiments found 8-37 % less erosion with vegetation (Figlus et al. 2014; Kosmalla et al. 2026), and intact cores of marram 36 % (De Battisti and Griffin 2020). The fine-root fraction of the traits was fitted to these, and the run falls in their range.

The model does not represent one effect seen in a large flume (Feagin et al. 2023): a vegetated dune that eroded faster, because plants changed the run-up and infiltration of the swash.

7 What remains uncertain

- The burial response and soil-biota pressure rest on few studies.
- Root reinforcement: the tensile strength of marram roots has not been measured, and the erosion data constrain only its product with the fine-root fraction.

Both uncertainties are listed with ranges in `marram_info`.

Bonte, Dries, Femke Batsleer, Sam Provoost, et al. 2021. “Biomorphogenic Feedbacks and the Spatial Organization of a Dominant Grass Steer Dune Development.” *Frontiers in Ecology and Evolution* 9: 761336. https://doi.org/10.3389/fevo.2021.761336.

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Feagin, Rusty A., Rachel A. Innocenti, Hailey Bond, et al. 2023. "Does Vegetation Accelerate Coastal Dune Erosion During Extreme Events?" *Science Advances* 9: eadg7135. https://doi.org/10.1126/sciadv.adg7135.

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Palmsten, Margaret L., and Robert A. Holman. 2012. "Laboratory Investigation of Dune Erosion Using Stereo Video." *Coastal Engineering* 60: 123–35. https://doi.org/10.1016/j.coastaleng.2011.09.003.

Walter, Benjamin, Christof Gromke, and Michael Lehning. 2012. "Shear-Stress Partitioning in Live Plant Canopies and Modifications to Raupach's Model." *Boundary-Layer Meteorology* 144: 217–41. https://doi.org/10.1007/s10546-012-9719-4.